

SMART INDIA HACKATHON 2025



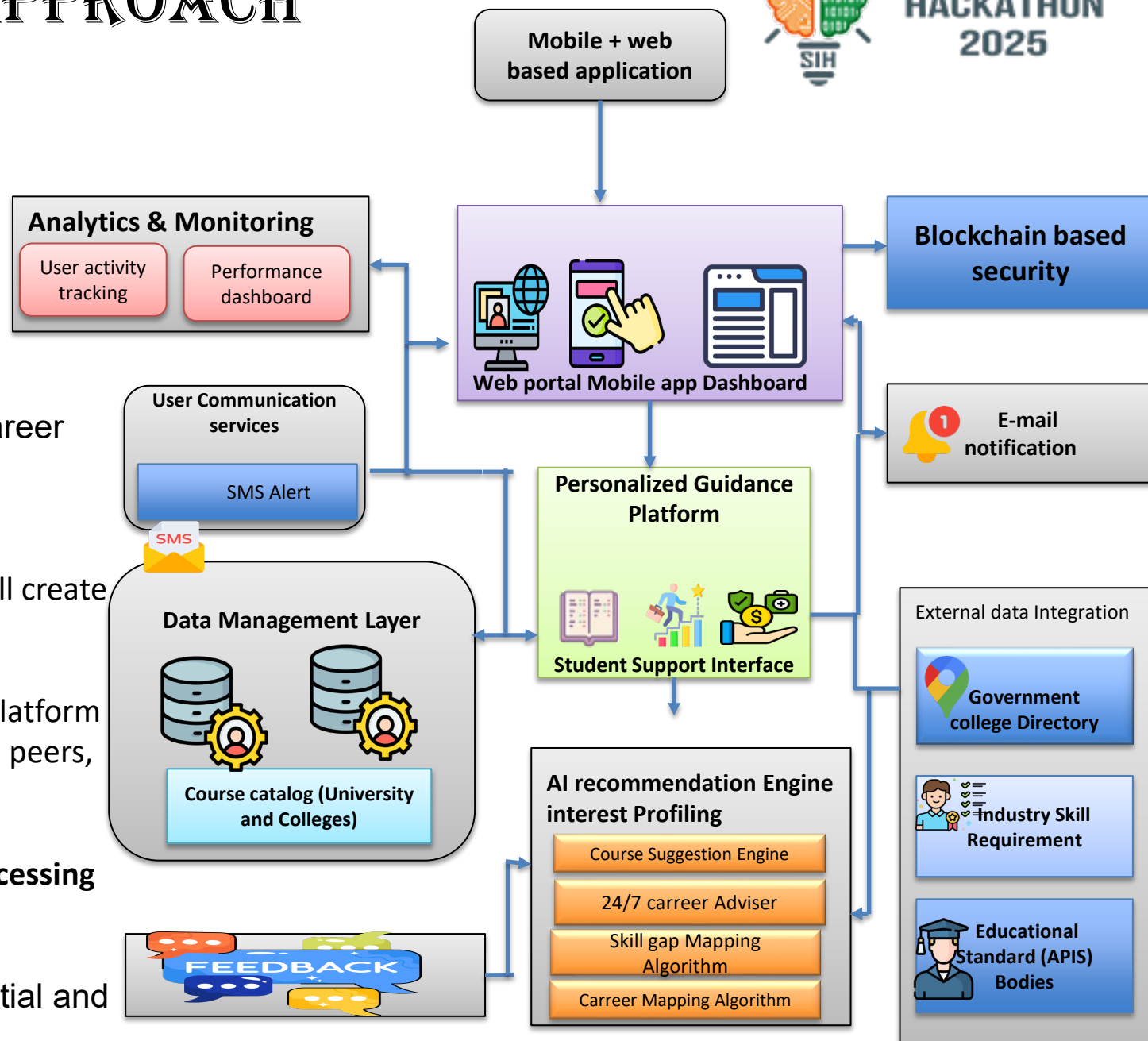
TITLE PAGE

- **Problem Statement ID –** 25094
- **Problem Statement Title-** One-Stop Personalized Career & Education Advisor
- **Theme-**Smart Education
- **PS Category-** Software
- **Team ID-**
- **Team Name (Registered on portal)-** MentEra



PROPOSED SOLUTION

- Comprehensive Guidance:** It will provide insights into the long-term outcomes of different courses, including potential jobs, entrance exams, and necessary skill development.
- AI-Powered Chatbot for Real-Time Guidance**
 An AI-Powered Chatbot acts as an instant, 24/7 virtual career and education guide, providing immediate answers and personalized advice to students.
- Local-Content-Driven Virtual Campus Tours:** our platform will create and upload their own free, 360-view of Campus.
- Community-Driven Q&A and Peer Mentorship Forum:** our platform will host a moderated forum where students can connect with peers, verified college students, and faculty.
- our platform integrates local language **Natural Language Processing (NLP)** to ensure accessibility and ease of use.
- Blockchain enabled technology:** Ensures user credential and security.



MentEra

TECHNICAL APPROACH



TECHNOLOGY STACK

Data Science: NumPy, Pandas(For Analysis), Matplotlib (For visualization).

AI/ML: Scikit-learn, ChatGPT, Gemini Pro, TensorFlow, machine learning.

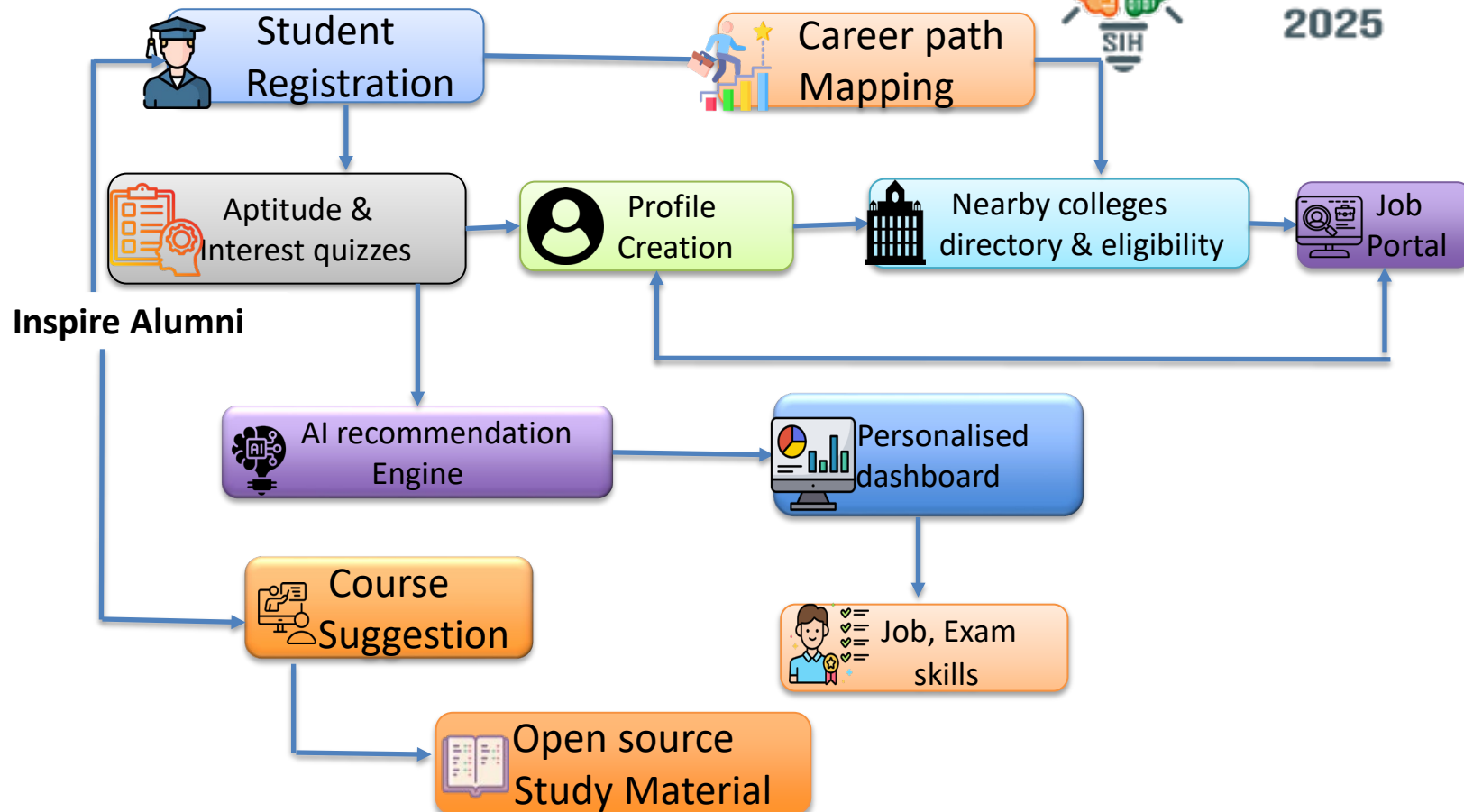
Frontend: ReactJS, React Native, Tailwind, Framer, Motion, HTML, CSS.

Backend: NodeJS, ExpressJS, REST API, MongoDB, Firebase, Solidity.

API Services: LinkedIn, GitHub, Gmail.



django



IMPLEMENTATION STRATEGY

Edtech and Revenue

Timeline Tracker

Stakeholder Development Partner

Pilot Launch (District/National)



FEASIBILITY

- 1. Technical:** Utilizes existing Technologies (Data Analysis, AI, Web, Blockchain).
- 2. Economic:** Low initial investment with Cloud based infrastructure .
- 3. Data Sourcing:** Our research indicates that government data portals like data.gov.in, AISHE etc.
- 4. Strong demand** for career and education adviser
- 5.** Our app would be a comprehensive **educational guidance platform**



VIABILITY

- 1. Sustainability / Long-term success** ✓: our platform keeps updating with career path , regular updates, new job trends etc.
- 2. Need in society** ✓: Many students (and even parents) are confused about what course or career to choose.
- 3. Competition:** Most platforms are either generic or exam-focused, Our platform is personalized, all-in-one platform and has good scope.
- 4. Trust & Credibility** ☀️ our platform will use **blockchain technology** Where every credential will be secure.

No. Of Schools

1471891

Urban

2.60 Lakh

Rural

12.12 Lakh

No. Of Teachers

9807600

Male

45.77 Lakh

Female

52.31 Lakh

No. Of Students

248045828

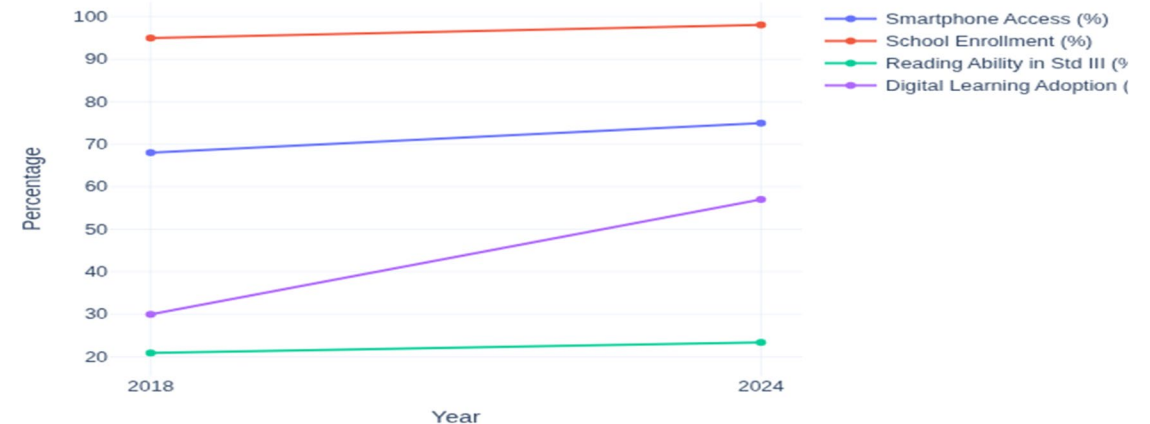
Boys

12.87 Crore

Girls

11.93 Crore

Improvements in Rural Education Indicators (2018 vs 2024)



Benefits of the Solution

Guide
student
Effectively

Reduced
Dropout

Human Capital
Development

Government
Resource
Saving



BENEFITS

1. MentEra helps to **Improve enrollment** in government degree colleges by helping students make informed academic decisions.
2. MentEra helps to **Reduced dropouts** after Class 10 and 12 .
3. Empowered students with access to reliable, **localized career guidance**.
4. **Stronger perception** of government colleges as viable career-building institutions.
5. **Optimized Public Investment** MentEra helps ensure that every rupee invested in education translates into productive human capital.



IMPACT

1. Empowers Informed Choices: Students from rural and underserved areas gain access to personalized career and education guidance, helping them make confident, informed decisions about their future.

2. Bridges Urban-Rural Divide: MentEra democratizes access to career counseling and educational resources, leveling the playing field for students in Tier-2, Tier-3 cities and rural regions.

3. Fosters Peer-Led Growth: Through mentorship forums, students connect with peers and alumni, creating a community-driven support system that enhances learning and motivation.

4. Scalable & Sustainable: Built on modern tech stacks and integrated with government data, MentEra is designed for long-term scalability and low-cost deployment across districts.

RESEARCH

1. Problem Understanding & Gap Analysis:

- Identified lack of unified platform for verified credentials, career guidance, and college discovery.
- Rural and Tier-2/3 students face limited access to mentorship and verified information.

2. Data Analysis:

- Used AISHE and UDISE+ to estimate student population and institutional coverage.
- NIRF rankings used to validate college quality and eligibility.

3. Technology Feasibility:

- ABC, NAD, Digi Locker provide official APIs for verified academic records.
- AI guidance aligned with UNESCO's GenAI in Education guidelines.

4. User Insights:

- Informal interviews with students and teachers in rural Punjab.
- High demand for local-language guidance, verified scholarships, and career mapping.

REFERENCES

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13. Hivebrite: <https://hivebrite.io>